UMAY — END-TO-END PIPELINE ARCHITECTURE & BIOPHYSICAL WORKFLOW
Autonomous Drug Discovery, Multi-Omics Target Prioritization, Modality Routing & Biophysical Design Suite

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SECURITY: **Confidential & Proprietary Biophysical Report**

1.0 END-TO-END 6-PHASE COMPUTATIONAL & BIOPHYSICAL PIPELINE OVERVIEW

PHASE 1: INPUT & CACHE	PHASE 2: STAGES 1-5 TARGET	PHASE 3: CENTRAL ROUTER	PHASE 4: STAGES 6-7 DESIGN	PHASE 5: STAGE 8 VALIDATION	PHASE 6: ENTERPRISE OUTPUT
Disease Query (EFO/MONDO)	**Genomics & 3D Cavitation**	**Autonomous Modality Decision**	**Bayesian Active Learning**	**Advanced Biophysical Polish**	**IND Dossier & Reporting**
MD5 SQLite api_cache.db + Neurological CNS vs Peripheral Classifier	Open Targets L2G • ClinVar • gnomAD v4.1 • DepMap • STRING • HPA • PDB/AF	Inhibitor • Activator/PAM (14Å Masking) • PROTAC (CRBN/VHL) • Gene Therapy	Mode 1 BRICS vs Mode 2 GRU RNN • 8D Vector • Cholesky GP • Vina MPO (Budget=32)	OpenMM 10ns MD • QM/MM PM6 • MM-GBSA Solvation • DiffDock • PLIP • Off-Target	st-mol WebGL • FastAPI 15 Endpoints • SHA-256 3-Page IND PDF Dossier • CSV

2.0 INPUT, GENOMIC CONSTRAINT & CENTRAL MODALITIES ROUTER MATRIX

Layer / Stage	Biological Mechanism & Algorithmic Method	Primary Data Source & Computational Engine	Acceptance / Gate Threshold
1. Input & Caching	Phenotypic ontology resolution and anatomical classification	EFO / MONDO API + SQLite (api_cache.db, TTL: 30 days)	MD5 Hash Hit (~1.2s Fast Response)
2. Stage 1: Genetic Evidence	Causal human GWAS loci and pathogenic clinical variants	Open Targets L2G, ClinVar, gnomAD v4.1 (pLI/LOEUF), DepMap Chronos	S_gen >= 0.15, pLI >= 0.90, Chronos < -0.5
3. Stage 2: Network Biology	Protein-protein interactome graph and bottleneck hub analysis	STRING DB v12 (Confidence >= 0.70), NetworkX Graph Topology	S_net >= 0.10, Degree & Betweenness Hub
4. Stage 3: Expression Safety	Vital organ on-target expression and systemic toxicity filtering	Human Protein Atlas (HPA) Consensus RNA, GTEx v8	S_safe >= 0.30, Low Cardiac/CNS Toxicity
5. Stage 4: Tractability	Small molecule, biologic, and PROTAC cavity druggability buckets	ChEMBL 33 Tractability, Pharos IDG Illumination	S_drug >= 0.10, Ligandable Cavity
6. Stage 5: 3D Structure & Cavity	Experimental crystallographic structure or AlphaFold 3D coordinates	RCSB PDB (X-Ray/Cryo-EM), AlphaFold DB, ESMFold, cKDTree	Resolution <= 2.5 Å, pLDDT >= 70
7. Router: Orthosteric Inhibitor	Silencing enzymatic hyperactivation and oncogenic drivers	ClinVar, UniProt, Target Phenotype Decision Tree	Active Catalytic Site Blockade
8. Router: Allosteric Activator	Restoring function in receptor desensitization and haploinsufficiency	14 Å Exclusion Sphere Cavity Masking & Allosteric Discovery	INS, INSR, GLP1R, CFTR, SCN1A Target Guard
9. Router: PROTAC Degrader	Targeted degradation of undruggable or resistant oncoproteins	CRBN / VHL E3 Ligase + 7 Covalent Linker Library	KRAS, MYC Stable Ternary Complex
10. Router: Gene Therapy	Recommending gene replacement for complete catalytic null alleles	AAV Viral Vector Capsid and Transgene Template	MPS IIIC Small-Molecule Synthesis Bypass

■■ MEDICAL PARADOX & ANATOMICAL PHARMACOLOGY TARGET GUARDS

1. **Metabolic & Hormonal Deficiency Guard:** In hormone/receptor deficiencies (e.g. INS, INSR, GLP1R, GCK in Type 2 Diabetes), orthosteric inhibition is permanently blocked; autonomous routing defaults to 'ACTIVATOR / PAM'.
2. **Dynamic Blood-Brain Barrier (log BB) Labeling:** In peripheral diseases, negative log BB is marked as an off-target safety asset ('Peripheral Restricted / CNS Safe'), eliminating unnecessary lipophilic optimization.
3. **PLIP Sequence Boundary Protection:** Amino acid coordinates are bounded to native polypeptide lengths; neutral amide bonds are strictly barred from false salt-bridge assignments.

3.0 STAGES 6–7: DUAL-MODE LEAD GENERATION & BAYESIAN CHOLESKY GP ACTIVE LEARNING

Component / Module	Algorithmic Mathematics & Methodology	Computational Framework & Parameters	Optimization & Convergence Goal
Candidate Mode 1	Virtual Library Screening	1000+ FDA/ChEMBL BRICS Fragment Pool	Approved Chemical Scaffold Diversity
Candidate Mode 2	De Novo Generative Design	10,000 SELFIES/SMILES GRU RNN Model	100% Chemical Validity, Strict Natal Filter
8D Feature Space	Multi-Dimensional Vectorization	[MW, LogP, TPSA, HBD, HBA, RotB, Fsp3, Arom]	Lipinski Ro5 & Veber Parameter Boundaries
Selector Brain (GP)	Gaussian Process Regression	Cholesky Covariance Factorization (K_XX = L * L^T)	Minimized Predictive Variance & Uncertainty
Acquisition Function	Lower Confidence Bound (LCB)	alpha(x) = mu(x) - kappa * sigma(x)	Exploration vs Exploitation Pareto Balance
MPO Multi-Objective	Multi-Parameter Objective Scoring	Vina Affinity + QED + SA + LE + BBB Penalty	Global Minimum Convergence within 32 Budget

4.0 STAGE 8: INDUSTRIAL BIOPHYSICAL VALIDATION & SAFETY PHARMACOLOGY

Validation Engine	Biophysical Principle & Foundation	Applied Software & Force Field Parameter	Acceptance & Safety Gate Threshold
Langevin MD (10 ns)	Classical Molecular Dynamics Simulation	OpenMM 8.1, AMBER ff14SB & GAFF2, 300K, 1 atm	RMSD < 2.0 Å, RMSF < 1.5 Å, Residence > 90%
QM/MM Refinement	Semi-Empirical Quantum Mechanics	PM6 Hamiltonian Level, HOMO/LUMO Band Gap	Enhanced Orbital Overlap & Dispersion
MM-GBSA Solvation	Continuum Solvation Free Energy	GB-OBC2 Dielectric + SASA Nonpolar Term	Delta G_bind < 0, Favorable Binding
DiffDock SE(3)	Equivariant Deep Learning Pose Check	SE(3) Equivariant Diffusion Model	Pose RMSD < 2.0 Å Cross-Validation
PLIP 3D Fingerprint	Structural Contact Profiler (Native Residues)	H-Bonds, pi-pi, sigma-hole Halogen, Salt Bridge	Residue Length Bounded, Neutral Amide Safe
Off-Target Shield	Safety Pharmacology Cross-Docking	hERG (QT Prolongation), CYP3A4, 5-HT2B Valvulopathy	Zero Safety Red Flags, Clear Cardiotox
Chemical Novelty	Morgan ECFP4 Fingerprint Screening	ChEMBL / PubChem 100k Benchmark Comparison	Max Tanimoto < 0.70 (High Novelty Tier)

5.0 ENTERPRISE DELIVERY LAYER, REST API & DIGITAL CRYPTOGRAPHIC SEAL

UMAY COMPUTATIONAL CHEMISTRY & STRUCTURAL BIOLOGY BOARD // CRYPTOGRAPHIC SEAL

System Title: UMAY — Unified Molecular Architecture & Yield Engine v2.0 Enterprise
REST Microservices: FastAPI REST v1 (15 Endpoints: /api/v1/health, /discover, /protac, /validation/stage8)
Visual Delivery: Streamlit WebGL st-mol / py3Dmol Interactive 3D Crystal & Cavity Inspection
Documentation: SHA-256 Cryptographic Sealed 3-Page IND PDF Dossier + 103-Parameter Master CSV
Regulatory Grounding: FDA CDER / EMA In Silico Preclinical IND-Enabling Guidelines Grounded